Assignment: Logistic Regression (Hands-on Exercise)

Please create a synthetic dataset and implement a logistic regression model end-to-end. The objective is to clearly understand each step and work with realistic data challenges.

Dataset Requirements

Your dataset should include:

- Numerical feature(s) suitable for logistic regression
- At least one categorical variable
- Missing values (introduced intentionally)
- A few outliers

Steps to Follow

1. Data Creation
- Generate a synthetic dataset based on a real-world scenario (e.g., predicting whether a customer churns, a patient has a condition, or a loan gets approved).

2. Exploratory Data Analysis (EDA)
- Visualize the data.
- Identify correlations, missing values, outliers, and relationships between variables.
- Note: Understand what to do when variables are highly correlated and implement the necessary changes.
- Check class balance and understand the implications of imbalanced data.

3. Data Preprocessing
- Handle missing values.
- Encode categorical variables.
- Treat or justify the handling of outliers.
- Scale numerical features where appropriate.
- Split the dataset into training and testing sets.

4. Model Building
- Implement a logistic regression model step by step.
- Understand the role of the sigmoid function and how probabilities are converted to class predictions.

5. Model Evaluation

- Evaluate the model using appropriate metrics (Accuracy, Precision, Recall, F1-Score, ROC-AUC).
- Interpret the confusion matrix and results.
- Understand when accuracy alone can be misleading.

6. Key Observations

- Summarize learnings from each step of the process.

Expectation

Focus on understanding and explanation at every stage. Clarity of reasoning is more important than complexity.